GeneGuide AI: AI-Based Prenatal Genetic Risk Assessment and Decision Support System

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Abstract

Understanding genetic risk usually requires a doctor to review family history, parent details and past pregnancy results one by one. This process is slow and difficult when counsellors are not nearby. This paper introduces GeneGuide AI, a web-based tool that helps estimate risk using information from parents, family and past pregnancies. We used a Random Forest classifier trained on a dataset of 1,250 records with 14 input features. GeneGuide AI places each case into one of three groups: Low, Moderate or High risk. The result is intended to be educational, not a diagnosis. Every result page clearly states this. The system is built with Python, Flask, SQLite, HTML, CSS and JavaScript. It includes user registration and login, a form with sections that check for correct input, and a history of past assessments for each user. On a test set of 250 records, the model achieved 85.2 percent accuracy. The F1-scores were 0.92 for Low, 0.87 for Moderate and 0.75 for High risk. Importantly, GeneGuide AI never mixed up Low with High risk. By using machine learning with an easy-to-use web interface, GeneGuide AI demonstrates how risk screening can become more accessible and easier to reach, while still encouraging users to talk to real doctors for final decisions. I hope GeneGuide AI will help families make decisions.

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